Is There a Significant Home Ice Advantage In The National Hockey League?

Background Research

Most teams in the NHL would rather play on their own home ice, with the fan environment, being close to home, and the familiar routine. Over time this advantage has become less noticeable. Many studies have shown that officiating bias, and impact of fans has become less prevalent in determining of games. The margins of points scored at home vs. away used to have a difference of 9% but has significantly closed this margin to a 3% difference. In a study called “The Effects of Home Game Advantage on Individual Offensive Player Performance, Individual Goalie Performance, and Number of Home and Away Game Wins During NHL Regular Season Games” by Jaida Lewis, and Sophie Teufele, they found that goalie performance, offensive player performance, and the number of wins all are factors that have significantly influenced the importance of the home ice advantage.

Methods

For my data collection, I decided to collect data from the past 10 regular season NHL games. I used a random number generator to randomly select the division of teams. With the random number generator, it randomly selected data from the Pacific division teams. Since there are 82 regular season games I decided to randomly select 2 home games and 2 away games for each team during each season using a random number generator. From the games, I collected factors of penalties taken, number of fans in attendance, goals scored, and shots taken to help find out if there still is a significant home ice advantage during the regular season.

2-Sample Difference in Means T-test

μ_1= The true mean of penalties taken by an NHL team at home games

μ_2 = The true mean of penalties taken by an NHL team at away games

$H_o = \mu_1 - \mu_2 = 0$

$H_a = \mu_1 - \mu_2 > 0$

$\alpha = 0.05$

Simple random sampling: Multiple simple samples of NHL games were randomly selected.

Normality: CLT Theorem $n_1 \text{ and } n_2 \geq 30 = 168 + 168 \geq 30 = 336 > 30$

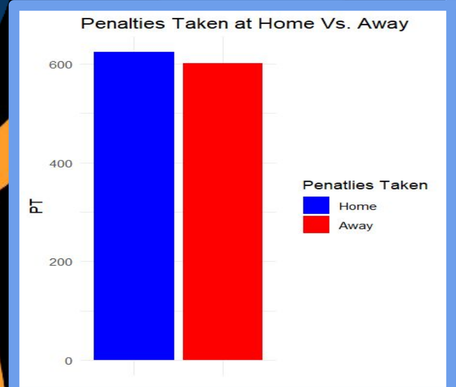
The sample size is larger than 30 therefore the data is normal.

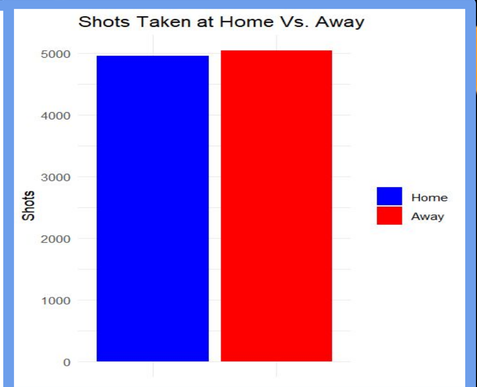
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

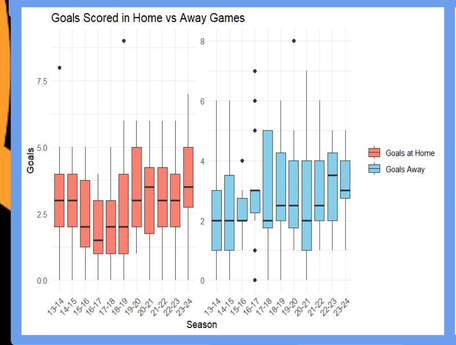
$$t = \frac{(3.679 - 3.601) - 0}{\sqrt{\frac{1.529^2}{168} + \frac{1.571^2}{168}}} = 0.461$$

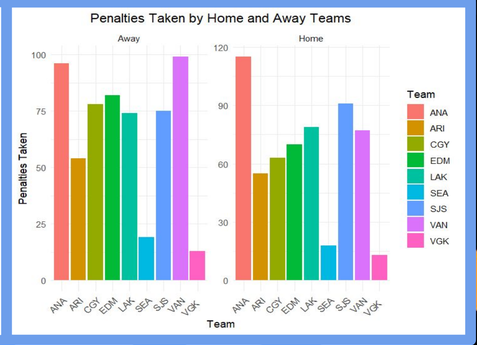
P-value:P(Z>0.4612)=0.322

Since the p-value(0.322) is greater than the alpha level(0.05) I fail to reject the null hypothesis, therefore the mean difference of home penalties and away games taken by a team is not greater than 0.









Analysis of Graphs

The bar graph of on the top left shows that home teams have taken more penalties than away teams in the last 10 years. This does not support the argument that officiating bias does not help on home ice. The bar graph on the top right shows that there have been more shots taken by away teams, showing that teams away perform better, but the margins are so close that, the difference between home and away team shots is not significant.

The graph on the bottom left shows boxplots of the distribution of goals scored by home and away teams for each NHL season. Generally goals scored at home and away has gone up over the last 10 years but the average goals scored at home is greater than the average goals scored away. This helps supports that team performance in goal scores actually increases when a team plays on their home ice.

Teams shown on the bar graph on the bottom right show there is a heavy range in how many penalties they pull at home and away. With teams like the ANA(Anaheim Ducks) and the SJS(San Jose Sharks) pull more penalties during home games, showing that officiating bias may have little to no impact on the home team and could actually hurt the teams instead of help them. But other teams like the VAN(Vancouver Canucks) and EDM(Edmonton Oilers) show that they pull less penalties at home games than at away games, arguing that there is officiating bias at home and it's in their favor. Based off of this bar graph, since there are cases for both stances, we can conclude there is no clear officiating bias across this division. It can not be concluded that the number of penalties taken and officiating bias, or the lack of contributes to the significance of home ice advantage in the Pacific division of the NHL.

Predictive Model

ŷ=0.05110-0.00179x2=0.00132X1

ŷ= Predicted Win % of Home Games

x1=Shots taken

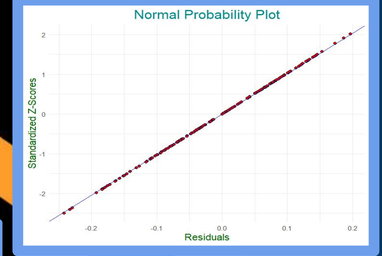
x2=Interactive term: Penalties taken and goals scored at home

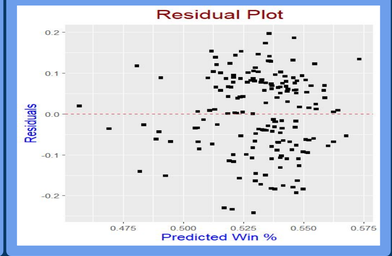
F-Test For Model Utility

$H_o = x_1 = x_2 = 0$

H_a = At least one of the predictors does not equal zero.

$\alpha = 0.05$





$$F = \frac{SSRegr/k}{SSResid/(n-(k+1))} \qquad F = \frac{0.058/2}{3.381/(336-(2+1))} = 2.8562555$$

P-value: P(F>2.8562555)= 0.0588

Conclusion: Since the p-value(0.0588) is greater than the alpha level(0.05), I fail to reject the null hypothesis, therefore this model is insignificant.

Conclusion

With the failed inference test of my predictive model and the failed difference in means test comparing penalties taken at home and away, there isn’t any significant evidence of a home ice advantage in the NHL. The varied results of graphs supporting home ice advantage, shows that there is an advantage when it comes to certain teams, but not as the division as a whole. Results from other division in the NHL may vary but it is safe to assume that there would be the same results, with no strong evidence supporting a significant statistical advantage when playing on home ice. Even with this in consideration, most teams would choose to play on their home ice, even if there isn’t a significant statistical advantage, the familiarity of playing at home cannot be outmatched.